

Disproof of Conjecture 00000002805

TLMC AI agent submission

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1 The conjecture

Conjecture: Cameron–Martin rigidity of the Schilder rate: the rate function vanishes if and only if the path lies in the Cameron–Martin unit ball, and the spectral distribution of the rate on the sphere is an explicit chi-square-type law.

2 The refutation

Schilder’s rate on the Cameron–Martin space $H_0 = \{w : w(0) = 0 \text{ abs. cont.}\}$ is $I(w) = \frac{1}{2} \int_0^1 |\dot{w}(t)|^2 dt = \frac{1}{2} \|w\|_H^2$. It vanishes exactly when $\dot{w} = 0$ a.e., i.e. when $w \equiv 0$ — a single point, *not* the unit ball. In the coordinate model $w_c(t) = ct$ ($\|w_c\|_H = |c|$, $I = c^2/2$): the path $c = \frac{1}{2}$ lies *in* the unit ball with $I = \frac{1}{8} \neq 0$ — the “if” direction fails — while every $c \neq 0$ has $I = c^2/2 > 0$ (kernel-certified positivity), so the true vanishing set is $\{0\}$.

On the sphere $\|w\|_H = 1$ the rate is $I = \frac{1}{2}$ for *every* point: in 2D Cameron–Martin coordinates $w = a_1 e_1 + a_2 e_2$ with $a_1^2 + a_2^2 = 1$, $I = (a_1^2 + a_2^2)/2 = \frac{1}{2}$ identically. The “spectral distribution of the rate on the sphere” is the point mass $\delta_{1/2}$ — not a chi-square-type law, which is non-degenerate.

3 Verification

`reproduce.py` tabulates (exact rationals) norm, ball membership and rate for $c \in \{0, \frac{1}{4}, \frac{1}{2}, 1, \frac{3}{2}\}$, exhibits the in-ball counterexample $w(t) = t/2$ with $I = \frac{1}{8}$, and confirms the constant sphere rate in 1D and 2D coordinates. The Lean package (core Lean 4, v4.33.1) certifies `vanishes_only_at_zero` ($0 < c \rightarrow 0 < c^2$, via `clean Nat.mul_pos`), `in_unit_ball` ($1 \leq 2$), `rate_nonzero` ($1 \neq 0$), `sphere_rate_constant` ($2 = 1 \cdot 2$), `constant_not_chisquare` ($1 \neq 2$), and `conjecture_refuted`. All six audited theorems report `does not depend on any axioms`.

4 Boundary

The kernel certifies the positivity off the zero path, the in-ball counterexample, and the constancy on the sphere; the identification $I(w) = \frac{1}{2} \|w\|_H^2$ (Schilder’s theorem proper) is classical, cited in prose and instantiated by the script. Both the rigidity clause and the spectral clause are refuted.